

Detection Capability and Generalization Boundaries of Interpretable Audio Phenomena

A Study of AI-Generated and Human Music

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Document status. This preserved working version adds the completed Native30 seven-family and BC-extension results, legacy spectral comparisons, and prediction-only YuE2 locked sensitivity to the earlier empirical chapters. The YuE2 aggregate report and Native60 frozen transfer are now included. Publication and full-corpus delivery reconciliation remain in progress; this is not a claim that every audio source has been redistributed. Earlier working versions are preserved separately.

Abstract

This thesis investigates interpretable spectral, dynamics, rhythm and structural measurements for distinguishing source-labelled AI-generated and human music. External controls test sensitivity to acoustic phenomena, while grouped, source-held-out and generator-held-out evaluations test distinct discrimination targets. On the expanded 4,228-recording Native30 development cohort, the spectral/dynamics/rhythm/phrase configuration achieves 69.72% group/source-balanced accuracy under generator holdout, compared with 66.29% for all eight evaluated families. The eight-family result is higher under ordinary grouped evaluation (78.37%), illustrating the importance of the evaluation target. A separate frozen-model Native60 transfer yields 91.70% mean AI sensitivity for the four families on 53 length-eligible locked YuE2 recordings, but cannot estimate human specificity. The study preserves failed attempts, intermediate outputs and reproducibility evidence. Its findings support measurable label associations, not an intrinsic or universally reliable acoustic signature of AI authorship.

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1 Research objective and scope

This thesis investigates the detection capability and generalization boundaries of interpretable audio phenomena in AI-generated and human music. Its objective is not to infer authorship from a single presumed acoustic signature. Instead, it asks whether explicitly defined measurements respond to their intended phenomena, provide incremental discrimination within a documented corpus, and retain that information when recording sources or generators are held out.

The motivating observation is perceptual: high-frequency characteristics, particularly in vocals, can influence a listener’s impression that music was generated. Turning this observation into a scientific claim requires more than a classifier trained on two collections. Frequency response, encoding, mastering, instrumentation, vocal activity and source separation can change the same measurements. A feature can be interpretable in its mathematical definition without uniquely identifying the physical process that produced an observed value.

1.1 Three distinct evidential questions

The study separates three questions. First, *measurement validity*: does a descriptor detect a controlled change in its intended phenomenon, and with what availability and nuisance sensitivity? Second, *discriminative utility*: does adding the descriptor improve a fixed classification procedure on matched recordings and evaluation splits? Third, *generalization*: does the resulting discrimination persist outside the recording sources and generators represented in training? These questions are related but not interchangeable. A passed external measurement gate does not establish classification utility; a high classification score does not retrospectively validate a failed measurement gate; and ordinary grouped evaluation does not establish unseen-generator attribution.

1.2 Operational meaning of the labels

Human and AI labels refer to the documented provenance of selected recordings. They do not label every musical decision or component in a production chain. Human-produced music may contain MIDI programming, synthesis, quantization, editing and compression. Consequently, regular dynamics or rhythm cannot alone establish AI generation. Conversely, a generated recording can exhibit variation in these quantities. The proposed measurements are statistical descriptors of audio, not necessary or sufficient conditions for either class. Provisional labels and small conditioning-group pilots are kept separate from validated accuracy claims.

1.3 Research questions

1. Which listening-motivated spectral differences remain observable after standardized audio preparation and an external separator-response correction, and what confounds remain unresolved?
2. Which additional phenomena can be measured reliably under the retained external controlled tests, including explicit negative controls and missingness accounting?
3. How do individual feature families and their nonempty combinations compare under fixed, source-aware splits and training-group caps?
4. Where do source-held-out and generator-held-out results diverge from ordinary grouped performance, and what does this imply for the scope of the detector’s claims?

1.4 Reporting commitments

All conclusions are tied to named cohorts, preprocessing versions and evaluation protocols. Recording identities, independent groups, audio views and stored files are different units and are reported separately. Comparisons across different duration cohorts are descriptive rather than causal estimates of duration effects. Missing measurements remain explicit; absence of a measurement is not silently interpreted as absence of the phenomenon.

The thesis retains unsuccessful candidates and negative feature increments. Numerical audits establish only the properties that they explicitly check; byte-level integrity is distinguished from waveform replay, and both are distinguished from external scientific validity. Raw classifier scores are not presented as calibrated probabilities. Existing development sweeps are not relabelled as untouched validation after inspecting their rankings.

The final experimental scope is restricted to the already collected material and the existing remaining analysis plan. New corpora, new phenomena and additional experimental branches are outside this thesis finalization run. Unresolved weaknesses are discussed as limitations and future work rather than addressed by unreported post-hoc experiments.

2 Related work and the limits of transferring evidence

The relevant literature spans several different target labels: synthesized speech, converted voices, lyrical authorship, source-separation fidelity and whole-recording music generation. An interpretable descriptor may be motivated by one task without being validated for another. This thesis therefore distinguishes the origin of a hypothesis from the external assay that qualifies its measurement and the held-source experiment that tests its predictive transfer.

2.1 Reconstruction artifacts versus generator transfer

Afchar, Meseguer-Brocal and Hennequin compare original music with matched autoencoder reconstructions, reducing content and file-encoding confounds [1]. Their controlled representation comparison reports 99.8% accuracy for amplitude and 99.6% for phase input. These are learned detector results on that reconstruction task, not validation of the present hand-designed F statistics. Their subsequent discussion of audio manipulation and unseen decoders illustrates why near-perfect internal discrimination is insufficient for deployment. We use this distinction to motivate both the original spectral/separator controls and the separate source-heldout evaluation; we do not reproduce their classifier or infer that its scores are comparable to our macro AUC or balanced accuracy.

2.2 Musical organization as measurement and representation

Wu and Yang analyze generated jazz lead sheets using pitch-class, groove, chord-progression and structural statistics, alongside listening and continuation tasks [9]. This provides a precedent for examining musical organization quantitatively. Its symbolic jazz setting does not establish a universal direction for chroma entropy or repetition in modern waveform generators. Our H and M implementations are explicitly specified audio descriptors, not replicas of every measure in that study.

Kim and Go’s Segment Transformer uses beat-aware segmentation and learned inter-segment relationships for music-generation detection [5]. The work evaluates short and full-audio systems on FakeMusicCaps and SONICS. It supports considering longer-context information, but a learned model’s improvement does not isolate a hand-designed recurrence scalar as its explanation. In our experiments, M’s standalone and incremental effects are measured separately, with duration eligibility preserved. The published model is neither retrained nor added to the current comparison.

2.3 Breathing evidence is task-specific

Mostaani and Magimai Doss study embeddings from networks trained to estimate speech breathing patterns in logical-access presentation attack detection [7]. Their ASVspoof 2019 results distinguish natural from synthesized speech, but do not show the same distinction for voice conversion. This is relevant motivation for investigating respiration-related cues, not validation of event-level breath localization in singing mixtures. The present project’s failed breath-event precision/F1 gate consequently remains informative even though earlier speech work reports successful discrimination. An embedding result and a breath-event counter have different measurement units and failure modes.

2.4 Synthesized stereo is not native-stereo music evidence

Liu et al.’s M2S-ADD first converts mono input to stereo using a pretrained synthesizer and then applies a dual-branch detector [6]. Their reported evaluation concerns ASVspoof2019. This does not establish that native stereo in generated songs has an intrinsic spatial inconsistency. Our SC family instead describes recorded channel relationships and must be interpreted alongside channel provenance, deliberately created width and the matched feature-ablation results. A classifier’s use of two channels does not alone identify the physical reason for its prediction.

2.5 Lyrics authorship and waveform generation

Frohmann et al.’s *Double Entendre* combines automatically transcribed sung lyrics with speech representations in a late-fusion system [3]. This provides a complementary approach using lyrical information, with robustness and generalization motivating its design. However, whether lyrics were generated and whether a recording was generated are not identical labels: human-written lyrics can condition synthesized audio, and generated lyrics can be performed by people. We do not treat lyrical evidence as ground truth for the present waveform-level labels or compare the paper’s reported performance directly with our source-heldout scores. No lyrics model is added in this closeout.

2.6 Position of the present experiments

The contribution examined here is not an assertion that one additional artifact universally separates human and generated music. It is the operationalization and testing of a finite set of interpretable descriptors, including explicit failures, availability limits and conditional predictive increments. The same external scientific gate, scalar definition and source-aware evaluation must be maintained even when a different choice would improve a displayed score. Literature-derived motivations cannot override observed failures of those tests.

The records above were rechecked against primary manuscripts or conference pages during editorial integration. They establish the stated task scopes; their experiments were not replicated here. The larger retained literature review and source-verification notes remain supporting provenance, not a claim that every proposed candidate entered this thesis’s experiments.

3 Data provenance and analysis populations

The project uses a dated recording registry, several overlapping analysis cohorts, and separate measurement-validation corpora. Their counts answer different questions. They must not be summed into a single number of independent songs, and external phenomenon-validation recordings are not automatically members of the AI/Human classification population.

3.1 Frozen expanded registry

The September 5 registry contains 10,141 selected recording identities: 3,091 Human and 7,050 AI-labelled. It has 10,140 distinct retained-native byte hashes and 3,979 global split groups. The duplicate MoisesDB pair shares a group and training weight. Exact-ten-second views cover the registry; 4,497 recordings satisfy the actual-duration condition for thirty-second analysis. Eligible development populations contain 8,361 and 3,317 rows, respectively. Locked, stress, provisional, and pilot roles are not silently admitted to development fitting.

Table 1: Human sources in the September 5 registry. Thirty-second eligibility does not imply admission to the development set. Roles describe that dated registry, not every subsequent experiment.

Source	Identities	Eligible 30s	Role
FMA	500	500	400 development, 100 locked
MTG-Jamendo	500	500	Development
DEAM	300	300	Locked
GTZAN	100	99	Stress
Hindustani HF	100	9	Stress
MAESTRO v3	300	300	Development
MagnaTagATune	500	0	Development
MedleyDB	178	168	Development
MoisesDB	239	239	Development
MusicNet	330	330	Locked
URMP	44	44	Development

The AI-labelled registry contains 500 entries each for ACE-Step, AudioLDM 2 Large, AudioLDM 2 Music, HeartMuLa, MusicGen Large, MusicGen Medium, MusicGen Small, Mustango, Riffusion, Stable Audio v1, Stable Audio v2, Suno, Udio, and provisional AudioX. DiffRhythm contributes 50 files but only ten conditioning groups. AudioX is not verified AI ground truth and receives no accuracy claim. Model-size or version variants are not assumed to be independent generator families. Short excerpts from some sources cannot support the long-window analyses merely because 500 files are available.

All 2,641 selected items in the materialization expansion were retained on disk and passed full-content SHA-256 validation. Their native, exact-ten-second, and unpadded up-to-sixty-second representations comprise 7,923 files and 83,210,585,185 bytes. This is a batch-specific storage total: archives, caches, stems, older data, model weights and subsequent acquisitions are additional. It is not a claim that the entire project occupies that amount of storage.

3.2 Matched duration cohorts

Equal60 v6 contains 2,174 exact-sixty-second recordings with native-stereo provenance: 1,278 Human and 896 AI. Its Human sources are MTG-Jamendo (481), MAESTRO (300), MedleyDB (156), MoisesDB (238), and Saraga (103). Its AI sources are Suno (396) and Mureka (500). The cohort excludes the 33 native-mono URMP recordings from the preceding v5 population. Mureka and Saraga had already entered exploratory development; their use here is not a fresh untouched confirmation.

Native30 contains 3,830 recordings, with 1,664 Human and 2,166 AI observations from eleven sources (Table 2). It expands generator coverage relative to Equal60, but remains restricted by duration and provenance eligibility. The cohorts overlap, and their sizes are not added to the older registry to manufacture an updated global census.

Native30 uses unpadded fixed excerpts and an explicit origin plan. A standardized 44.1-kHz file does not establish that the original recording possessed that bandwidth: native sample-rate

Table 2: Frozen Native30 cohort. All input views have exactly 1,323,000 frames at 44.1 kHz, two channels, and WAV FLOAT storage.

Human source	N	AI source	N
FMA	354	ACE-Step	400
MTG-Jamendo	500	HeartMuLa	366
MAESTRO	300	Suno	400
MedleyDB	168	Udio	500
MoisesDB	239	Mureka v9	500
Saraga	103		
Total Human	1,664	Total AI	2,166

eligibility remains joined to original provenance. Likewise, stereo measurements require meaningful native-stereo eligibility, not duplicated mono channels. Internal mono 16-kHz analysis views used by some measurements do not replace the stereo source needed by SC.

3.3 Measurement availability is not cohort membership

The committed Native30 FHSC measurement package retains all 3,830 rows. Complete selected scalars are available on 3,819 rows for F, 3,789 for H, and 3,829 for SC. F has 3,829 rows with status `ok`, but that status does not imply that every one of its fifteen selected scalars is observed. M is short-duration unavailable throughout Native30 and is not a primary predictor there. Thus Equal60’s eight families include M, whereas Native30’s seven-family base excludes M; BC is the eighth family in the completed extension reported later in this thesis.

The FHSC integrity audit verified 7,663 committed products, including 3,830 receipts and 3,830 SC frame archives. It established declared byte, schema, lineage and missingness properties, not independent waveform re-extraction of all features. Scientific missingness is retained rather than used to select a more favorable cohort.

The seven-family classification input package was subsequently assembled and verified on September 11, retaining the same 3,830 observations. Counting a row as complete only when every selected scalar in a family is finite gives S: 3,830; D: 3,830; R: 3,739; P: 465; F: 3,819; H: 3,789; and SC: 3,829. These counts are not interchangeable with producer eligibility flags: in particular, eligibility for some section measurements does not guarantee observability of every duration- and bar-based P feature. Missing predictors do not cause recording removal from the comparison cohort; preprocessing remains training-only. Package completion establishes a joined input table, not successful classifier fitting or improved discrimination. The package COMMIT is 361d499824cf1340ef5b65073685bbf2b981002545ee80485fe07374e1607863.

3.4 Data and audit references

The dated registry and physical materialization evidence are retained under `artifacts/source_diversity_expansion_20260905/`, specifically `results/inventory/corpus_inventory.md` and `audit/materialization_validation.json`. Native30’s committed cohort is identified by its contract hash 730b9bb25d18715c5ed0191ec9eb7a921d473b9534c9e5bc9196c25f5ce44510. Its source plan and selected-scalar availability audit are retained under `artifacts/audio_phenomena_expansion_20260907/audit/` as `native30_origin_plan_v2.json` and `native30_fhsc_committed_independent_audit_v1/REPORT_EN.md`. The final reproducibility appendix records full local and NFS locations; the local artifact file count is not substituted for a music-corpus count.

4 Evaluation targets and limits of interpretation

This thesis separates three questions that are often conflated in audio authenticity research. First, does a measurement respond to its intended audio phenomenon under controlled conditions? Second, does it discriminate the labels in a specified collection? Third, does its utility survive a change in recording source or generator? A positive answer to one question does not establish the others. In particular, a classifier may exploit production or acquisition conditions without identifying a consequence of generative synthesis.

4.1 Labels and the unit of analysis

The Human label denotes supported recording provenance, not an absence of MIDI, sampling, quantization, pitch correction, compression, or editing. Conversely, an AI-labelled recording may depend on human prompts, musical conditioning, or post-production. Dynamics and timing statistics therefore cannot by themselves establish whether a performance used MIDI or whether its authorship was human.

Recording identities, byte-identical audio objects, excerpts, derived stems, and split groups are distinct units. Multiple views of one recording do not provide independent evidence. Global groups preserve known dependencies across sources and, where applicable, across labels. Cohort totals from different duration experiments overlap and are not added to form a corpus size. Unverified labels and pilot conditions are reported separately from eligible classification populations.

4.2 Duration and measurement availability

Short recordings are not padded to create apparent evidence of long-term structure. An unavailable beat sequence or an insufficient section observation is not evidence of perfectly constant rhythm or zero structural variety. Natural unobservability, processing failure, and an observed numerical zero are retained as different states. A completed feature row is also not equivalent to complete observation of every scalar in that row.

The exact-duration experiments control the physical analysis window within each cohort. They do not isolate a causal effect of duration when their eligible populations differ. Similarly, native-stereo eligibility narrows the population to which stereo findings apply; duplicating mono channels does not manufacture spatial information.

4.3 Preprocessing, fitting, and decision scores

Imputation and scaling are fitted on training observations only. The evaluation retains the specified class, source, group, and recording weights, rather than allowing large catalogues or duplicate views to dominate the objective. Training quantity caps refer to global groups per source, not a universal number of songs. Small sources are not duplicated to satisfy a nominal cap.

The frozen tabular classifier uses a ridge penalty of 10, an unpenalized intercept, and a fixed decision threshold of 0.5. Its raw scores are unbounded and are not calibrated posterior probabilities. Consequently, a score of 0.8 must not be described as an 80% probability of AI generation. Missingness-only and related diagnostics test selected alternative explanations; they do not establish that every feature combination is free of processing artifacts.

4.4 Separate evaluation directions

Human-source-held-out evaluation examines transfer to an excluded Human source; generator-held-out evaluation examines transfer to an excluded AI source or declared generator family.

Ordinary grouped evaluation is reported separately. The latter can be informative about interpolation among observed sources while remaining optimistic about an unseen generator. Results from these directions are not interchangeable estimates of one deployment accuracy.

For studies using the composite criterion, let A_H and A_G denote the specified source-macro AUC in the two transfer directions. Then

$$J = \frac{1}{2}(A_H + A_G).$$

This is a mean of ranking statistics, not classification accuracy. Balanced accuracy at the frozen threshold is reported separately, with the same declared aggregation order. Scores produced by different fitted models are not casually pooled into a cross-model AUC. Source-specific sensitivity and specificity remain necessary even when a macro summary is favorable.

4.5 Feature and training-quantity ablations

Nonempty feature subsets and the five declared training caps are retained, including negative increments. A useful standalone measurement can be redundant or harmful in a larger model. Matched add-on contrasts therefore hold the cohort, folds, quantity policy, preprocessing, weights, and threshold fixed wherever the protocol specifies such a comparison. A maximum observed over many development configurations is a descriptive result, not an independently validated winner. Changes in sample count, weighting, and effective regularization also prevent a training-cap contrast from being interpreted as a pure causal effect of data diversity.

4.6 Uncertainty and evidence scope

Conditional resampling of held-out recording groups describes uncertainty within fixed fitted models and observed recordings. It does not include retraining or model-selection variability. Separate resampling of observed source labels characterizes sensitivity to that source composition; it does not turn a purposively assembled collection into an independent sample of future music domains. Undefined cells and the declared aggregation policy are retained rather than replaced by zero or chance performance.

Software tests, byte hashes, schema audits, independent numerical replay, controlled interventions, and external classification answer different questions. A hash confirms identity within its stated scope; it does not prove scientific validity. Numerical replay verifies selected arithmetic, not a causal account of authorship. A controlled phase-coupling response, for example, establishes a measurement capability under those conditions, not an AI/Human classification result. Failed admission gates and original failed audits are retained alongside any justified implementation corrections.

4.7 Reproducibility references

The authoritative implementations and result-specific protocols are indexed in the reproducibility appendix. Relevant preserved implementations include `extract_native30_sdrp_operational_v1.py`, `evaluate_native30_v3.py`, and `bootstrap_sdrfh10_uncertainty.py`. Their existence is not asserted as proof that an associated production run completed; empirical chapters require the corresponding committed outputs and verification evidence.

5 Spectral motivation, separator response, and source expansion

The initial hypothesis concerned audible high-frequency characteristics, especially in vocals. Candidate measurements represented band-energy ratios, spectral tilt, entropy, flatness, crest,

sibilance-related contrasts and bursts, and spectral residual structure. They are descriptions of spectral distribution, not direct measurements of perceived naturalness. Neither high nor low high-frequency energy is a necessary condition for generated music.

An early metadata negative control reached AUC 1.000 and BA 0.967 using original container-related information. This demonstrated that the initial collections could be separated for reasons unrelated to the proposed acoustic mechanism. High exploratory audio scores were therefore not accepted as evidence of universal authorship detection. Standardized views, source-aware evaluation and retained codec/low-pass controls became necessary parts of the study.

5.1 Separating the vocal hypothesis from separator error

The work progressed from mixed audio to separated-vocal analysis through stem pilots, a 100+100 comparison and expanded 500+500 spectral-band experiments. Neural source separation used RTX 5090 hardware. A separated vocal is an estimate, not an oracle reference; leakage and reconstruction error can affect the same high-frequency measurements that motivated the hypothesis.

The MUSDB18-7 oracle comparison processed 144 excerpts, including fifty official test tracks. It used a 4,096-point Hann STFT, hop 1,024, 44.1-kHz audio, a stereo-power magnitude representation, least-squares gain alignment and an activity floor of -60 dB. Signed error was estimated-vocal dB minus oracle-vocal dB. Frequency/error distributions and representative and severe STFT-error heatmaps were retained. On the fifty test tracks, median SDR was 8.76 dB and median per-track absolute STFT error approximately 3.28 dB.

An external average frequency-dependent response $b(f)$ was then estimated without AI/Human labels. The correction applied

$$|\widehat{V}_{\text{corr}}(f, t)| = |\widehat{V}(f, t)| 10^{-b(f)/20},$$

retaining the estimated phase. This is a multiplicative response correction in the magnitude domain, equivalent to a dB-domain offset. It is not the subtraction of time-aligned waveforms from unrelated songs, nor a general recovery of separator errors.

Held-out MUSDB median absolute-bin error improved from 3.276 to 2.835 dB. However, on one independent lossless MedleyDB vocal recording, error increased from 0.296 to 2.144 dB. That single recording cannot estimate a population failure rate, but it contradicts a claim of uniformly beneficial correction. The correction is interpreted as a MUSDB-conditioned stress test.

Table 3: Historical 500+500 locked comparison before and after external separator-response correction. These are not Native30 or unseen-generator results. Cells give BA / AUC.

Representation	Raw	Corrected
22 scalar metrics	0.740 / 0.785	0.755 / 0.794
0.3–20 kHz band vector	0.835 / 0.879	0.840 / 0.876
Band vector excluding 5–10 kHz	0.845 / 0.922	0.855 / 0.922

The shared average response did not remove most observed discrimination (Table 3). This does not exclude class-dependent separator errors, genre, mastering or codec explanations. A single air-ratio descriptor had AUC 0.637 in the corrected expanded set, compared with 0.744 in the earlier pilot; this descriptive cross-version contrast cautions against extrapolating small-sample effects.

5.2 Generator-specific spectral differences

The open-model generation work retained 500-prompt experiments for HeartMuLa and ACE-Step, including prompt and generation provenance. Muse supplied text prompts; its Suno waveforms were not used as Human recordings. Historical corrected locked band results were AUC/BA 0.922/0.855 for Suno, 1.000/1.000 for HeartMuLa and 0.988/0.970 for ACE-Step. Each was a generator-specific task with its own fitted model, not one detector evaluated on all unseen generators.

Corrected AI-minus-Human mean differences in the 10–16-kHz region were +4.322 dB for Suno, +14.104 dB for HeartMuLa and +8.465 dB for ACE-Step. Such heterogeneity motivates generator-held-out evaluation rather than a single presumed high-frequency AI template. The label-specific spectral distributions also remain vulnerable to production and collection confounds.

5.3 Expanded four-family development ablations

The expanded S/D/R/P study evaluated all fifteen nonempty subsets at caps 25, 50, 100, 200 and all global groups per source. The per-record spectral table already covered 7,100 development inputs before the later materialized expansion; it was not restricted to the original 1,500 AI recordings. The external MUSDB correction remained fixed rather than being re-estimated from classification labels. The primary S8 representation restricted analysis below 7.5 kHz with native-rate eligibility; full-band S16 was a separate sensitivity.

Table 4: Historical expanded four-family source-transfer results at the all cap. J and BA are separate macro estimands. Different duration populations prevent a causal duration interpretation.

Set	10s J	10s BA	30s J	30s BA
S	0.6191	0.5903	0.7186	0.6611
D	0.5487	0.5430	0.6698	0.6370
R	0.4791	0.4662	0.6566	0.6084
P	—	—	0.5030	0.5153
S+D	0.6334	0.5971	0.7332	0.6711
S+D+R	0.6272	0.5931	0.7641	0.6986
S+D+R+P	—	—	0.7601	0.6942

P was wholly unavailable at ten seconds; combinations containing it were ineligible for selection. Thirty-second P duration metrics were available on only 753 rows, with complete bar-span metrics on 414. Missingness was not interpreted as a measured lack of structural variation.

At caps 25, 50, 100, 200 and all, the ten-second S+D criterion was 0.6198, 0.6226, 0.6297, 0.6317 and 0.6334, respectively. The corresponding thirty-second S+D+R values were 0.7516, 0.7591, 0.7613, 0.7614 and 0.7641. Gains flattened at larger caps, but changing counts, weights and effective regularization prevents attribution to data quantity alone. Small differences were not supported by CV intervals.

Locked Human specificity exposed a stronger limitation: the thirty-second candidate reached 95.15% on MusicNet (330 recordings), but only 52.33% on DEAM (300 recordings), corresponding to a 47.67% DEAM false-positive rate. Udio family-held-out AUC was 0.5479 and sensitivity 0.1586. These failures preclude a reliable general-purpose attribution claim despite favorable within-source spectral separation.

5.4 Source and output references

Retained report roots under `artifacts/` are `frequency_ablation_20260813/REPORT.md`, `musdb18_oracle_vocal_eval_20260901/analysis/REPORT.md`,

demucs_bias_corrected_1000_20260901/comparison/REPORT.md, open_models_spectral_500_20260901/analysis/comparison/REPORT.md, and source_diversity_expansion_20260905/RESULTS_SUMMARY.md. The expanded full fifteen-by-five tables and their numerical audits are in the latter experiment's results/verified_presentation/ and audit/formal_results_consistency.json. These historical results remain separate from the finalized Exact10 five-family and matched Equal60 analyses.

6 Operational definitions of the original four families

The names S, D, R and P denote fixed measurement families, not four independently validated authorship mechanisms. The expanded extractor uses Demucs vocal estimates for S, the sum of bass/drums/other estimates for D, retained beat/downbeat estimates for R, and retained All-In-One segment boundaries combined with downbeats for P. Neural estimates are inputs to deterministic measurements; they are not ground-truth events.

The retained neural front ends are Hybrid Transformer Demucs [8], Beat This! [2], and All-In-One metrical and functional structure analysis [4]. The actual checkpoint manifest resolves `htdemucs` to signature `955717e8`, Beat This! to `final0`, and `harmonix-all` to the ordered eight-model ensemble `harmonix-fold0` through `harmonix-fold7`. Thus the separator citation is for the Hybrid Transformer version, not merely its earlier hybrid predecessor; `htdemucs_ft` is not silently substituted. The manifest at `artifacts/source_diversity_expansion_20260905/results/inference_checkpoint_manifest.json` retains the resolved file hashes. Native30 binds that manifest in its inference contract. Published front-end benchmark scores do not establish the accuracy of their event estimates on this project's generated music.

6.1 S: bandwidth-controlled vocal spectral statistics

The expanded implementation uses mono float64 vocal estimates with DC removal, 44,100 Hz, 4,096-sample frames and hop 1,024. Its explicit `numpy.hanning` window is symmetric; this differs from the periodic Hann used in several other modules and must not be silently interchanged when reproducing results. Power receives an epsilon of 10^{-14} before the frozen external frequency-response correction. Active frames exceed the larger of the 30th-percentile RMS level and the maximum minus 35 dB; if fewer than 16 qualify, the strongest available frames are selected up to 16. Consequently, frame selection alone does not certify audible vocals. A separately retained vocal-activity sensitivity requires vocal-to-mix RMS at least -18 dB, vocal RMS at least -60 dBFS and 16 active frames.

S8 denotes a common-band representation, not eight descriptors: it has 15 values and principally uses the 5–7.5 kHz interval, with lower-frequency reference bands. S16 has 16 values, including a 12–20 kHz air-energy ratio, and is a separate sensitivity representation. Original native sample-rate eligibility is at least 16 kHz for S8 and 40 kHz for S16; resampling cannot restore absent source bandwidth. The Native30 primary S family retains S8.

For corrected frame power $P_{k,t}$, high-band flatness is geometric mean divided by arithmetic mean; normalized spectral entropy is $-\sum_k p_{k,t} \log p_{k,t} / \log K$; crest is the maximum-to-mean power ratio expressed in dB, with implementation epsilons retained. Energy ratios compare high-band power with 1–5 kHz body power and sibilance-band power with 0.3–5 kHz power. These frame statistics are summarized by medians. Spectral slopes fit mean log power against $\log_2(f)$ and therefore have units dB/octave.

The remaining descriptors capture spectral-envelope residual peaks and periodicity, adjacent active-frame log-spectral cosine similarity and its complement, high-band power standard deviation, a nominal 4–12 Hz modulation share, sibilance contrast and burst rate. Residual peaks use height 0.35, prominence 0.08 and minimum two-bin spacing after the fixed lower-envelope nor-

malization. Contrast is the 90th percentile minus median of the sibilance ratio. Bursts exceed median plus 1.5 MAD; their rising-edge count is divided by the selected-frame duration.

An important implementation limitation is that active frames are selected *before* adjacent-frame and modulation calculations. The latter use the nominal hop-derived frame rate on this compressed active-frame sequence. Thus, when activity contains gaps, adjacency need not mean neighbouring waveform frames, and nominal modulation Hz and burst rates are not exact continuous-time event measurements. These frozen descriptors are retained for reproducibility, not relabelled as validated physical modulation rates. The separate Q external assay must not be conflated with these S descriptors.

6.2 D: event-conditioned accompaniment amplitude dispersion

The accompaniment sum is converted to mono float32. A 4,096-point Hann STFT with hop 512 and no boundary/tail padding supplies 30–8,000 Hz power. Positive differences of $\log(1 + 10^6 P)$, averaged across selected bins, form the onset-strength sequence. Peaks must exceed median plus one MAD and be at least approximately 80 ms apart.

For each candidate, the median power profile over 20–120 ms after the STFT peak time is formed. Up to the 16 strongest local spectral maxima contribute summed energy, expressed as $10 \log_{10}(E + 10^{-12})$. With at least eight valid event amplitudes, D reports the 90th-minus-10th percentile span, IQR, and median absolute adjacent-event change, all on this logarithmic amplitude scale. Otherwise the values remain missing. This is pseudo-velocity dispersion of an estimated accompaniment mixture, not instrument-separated MIDI velocity or proof of MIDI control. Different instruments, overlapping attacks, compression and separation errors can alter the event distribution.

6.3 R: beat-interval and local-tempo variation

R requires at least 16 beat estimates and at least 15 positive finite inter-beat intervals (IBIs). Despite its historical name `ibi_cv`, the statistic is $\text{IQR}(\text{IBI})/\text{median}(\text{IBI})$, *not* the conventional standard-deviation coefficient of variation. Instantaneous tempo is $60/\text{IBI}$; values outside 30–300 BPM are removed. With at least 14 remaining values, `tempo_tv` is median absolute adjacent tempo change divided by median tempo. Entropy uses five-BPM histogram bins from 30 to 300 and normalization by the logarithm of the full bin count, not only occupied bins.

Removing invalid intervals/tempos can create adjacency across an excluded value; this retained statistic is a robust sequence descriptor rather than an exact time-integrated tempo derivative. Empty valid beat output and missing or malformed inference output have different provenance states. Neither may be interpreted as an observed perfectly constant rhythm.

6.4 P: segment-duration and downbeat-span proxies

All-In-One segment starts and the final segment end are clipped to the view, deduplicated and cleaned with a minimum one-second separation; view endpoints zero and duration are included. At least three positive segment durations are required for median duration, IQR/median and normalized two-second-bin duration entropy. Histogram support is at least 32 seconds and expands for longer segments. The historical `section_duration_cv` name again denotes IQR/median .

With at least four downbeats and four boundaries, boundaries are snapped to nearest downbeat indices and deduplicated. Three positive index spans are needed for median span, IQR/median and the fraction unequal to the modal span; ties select the smallest modal span. These are estimated downbeat spans, not annotated phrase lengths. Boundary snapping and endpoint truncation may change them substantially, especially in short views. The failed external P gate and missingness are reported separately.

6.5 Version and reproduction boundaries

The definitions above follow `artifacts/source_diversity_expansion_20260905/code/expanded_feature_definitions.py` and `code/extract_expanded_four_family.py` in that same directory. Native30 binds this backend through `artifacts/audio_phenomena_expansion_20260907/code/extract_native30_sdrp_operational_v1.py` and its frozen dependencies. The older decoder permits limited short-tail padding; this must not be described as a universal no-padding guarantee. Native30 separately verifies exact physical input and inference-output support before using the backend. Historical pilots, S16 sensitivities and earlier band-vector models are distinct representations, as indicated in their respective result tables. No definition or threshold was changed while writing this section.

7 Additional deterministic measurement definitions

The additional families are operational descriptors, not interchangeable definitions of a generation artifact. Their scalar catalogues are fixed separately from quality fields and diagnostic arrays. A missing measurement is not a measured absence of the phenomenon. Native30 uses 15 F, six H and six SC descriptors in its 54-column seven-family representation; the BC extension appends exactly one scalar. Equal60 additionally uses six M descriptors. The following definitions describe retained implementations, not themselves evidence of classification performance; the completed Native30 results are reported separately below.

7.1 F: phase residuals and local group-delay dispersion

For STFT phase $\phi_{k,t}$, frequency f_k , sample rate f_s and hop h , the phase residual is

$$r_{k,t} = \text{wrap}(\phi_{k,t+1} - \phi_{k,t} - 2\pi f_k h / f_s). \quad (1)$$

With valid-pair weights $w_{k,t} = \sqrt{|X_{k,t+1}| |X_{k,t}|}$, the reported circular variance is

$$V_r = 1 - \frac{\left| \sum_{(k,t) \in \mathcal{V}} w_{k,t} e^{ir_{k,t}} \right|}{\sum_{(k,t) \in \mathcal{V}} w_{k,t}}. \quad (2)$$

Aggregation is joint across valid time–frequency pairs, not a mean of independent per-bin circular variances. The expected rotation has the forward-FFT sign convention used by the implementation.

At 16 kHz the window is 1,024 samples (64 ms), with a 256-sample hop, periodic Hann weighting, no STFT boundary extension and no tail padding. The module subtracts the waveform mean. Its absolute waveform RMS floor is -100 dBFS. Bin magnitude must exceed both the frame maximum minus 40 dB and the excerpt maximum minus 70 dB; frequencies are restricted to 50 Hz through 95% of Nyquist. Active frames use a separate relative frame-power rule: the peak minus 50 dB when peak-to-20th-percentile spread is below 12 dB, otherwise the peak minus 45 dB.

The implemented local delay proxy is

$$\tau_{k,t} = -\frac{\text{wrap}(\phi_{k+1,t} - \phi_{k,t})}{2\pi/N} \frac{1000}{f_s} \text{ ms}. \quad (3)$$

It is a wrapped adjacent-bin phase-difference proxy, not an unwrapped physical propagation-delay estimator. Reliable adjacent bins are required. Within each frame, the median of within-band delay IQRs and the IQR of band-median delays are computed. The bands at 16 kHz are 50–500, 500–2,000, 2,000–4,000 and 4,000–7,600 Hz. A band needs at least three valid delay values; cross-band dispersion needs two band centres.

All-frame, attack, sustain and decay summaries are retained. Regions are assigned from the slope of a three-frame-smoothed log-energy sequence, using its active-frame median and a threshold of $\max(0.5, 2.5 \text{ MAD})$. These are energy-direction regions, not annotated instrument envelopes. The all-frame region requires eight active frames and other regions require six; circular variance additionally requires 16 valid pairs. Delay summaries need at least $\max(3, \lfloor n_{\text{region}}/4 \rfloor)$ valid frame values. Three decay-minus-sustain contrasts complete the 15 features.

7.2 H and M: chroma paths and bounded-lag recurrence

The chroma implementation uses a 4,096-point Hann STFT at 16 kHz, a 1,024-sample hop, no centring, power input and fixed tuning zero. Frames are active above 1% of the maximum spectral RMS. L1-normalized chroma must have normalized entropy at most 0.92 to be informative. One-second blocks average informative chroma frames and are then L2-normalized; each block needs at least two informative frames and at least 25% of the rounded expected frame count.

H requires at least eight seconds and the implementation’s block/path coverage checks. Its six descriptors are normalized aggregate pitch-class entropy, dominant-token entropy, median and IQR of adjacent-block cosine distance, median circular dominant-token step, and the rate of large steps. For dominant pitch-class tokens $a, b \in \{0, \dots, 11\}$ the step is $\min(|a - b|, 12 - |a - b|)/6$; a large step is at least 0.5. Only adjacent eligible blocks are compared: a gap is not bridged to create a transition. These quantities do not constitute chord transcription or harmonic quality.

M requires at least 45 seconds. Four-second chroma patterns are compared at integer lags from eight seconds through $\min(60, T - 4)$, allowing all 12 pitch-class rotations. The implementation does not build a full self-similarity matrix. For each lag it retains the 90th-percentile similarity and the fraction at or above 0.88. It selects an early sufficiently supported lag within 0.01 of the global peak, with the specified density safeguard. The six outputs describe peak similarity, density, lag contrast, selected lag, transposition and returning-pattern count. They are not independent motif events. Native30 cannot meet the 45-second requirement, so M is not a Native30 primary family.

7.3 SC: retained stereo descriptors

The primary SC catalogue selects three bands, 80–500, 500–2,000 and 2,000–6,000 Hz, with two descriptors per band: median absolute interchannel level difference and median side-energy fraction. For complex band coefficients L and R , mid and side are $(L + R)/2$ and $(L - R)/2$; the latter descriptor is

$$q_{\text{side}} = \frac{\sum_k |(L_k - R_k)/2|^2}{\sum_k |(L_k + R_k)/2|^2 + \sum_k |(L_k - R_k)/2|^2}. \quad (4)$$

This fraction lies in $[0, 1]$ on eligible frames. Silent bands are masked; interchannel level difference additionally requires sufficient energy in both channels. Full coherence, interchannel-phase and five-band arrays remain diagnostic evidence, not extra primary predictors. These measures may capture panning and mastering conventions; they are not intrinsic authorship signatures.

7.4 BC: fixed-triplet squared bicoherence

The fixed target is $(f_1, f_2, f_1 + f_2) = (500, 750, 1250)$ Hz. Within one four-second pool, let $A_k = X_k(32)X_k(48)$ and $C_k = X_k(80)$. Then

$$b^2 = \frac{|\sum_k A_k \overline{C_k}|^2}{(\sum_k |A_k|^2)(\sum_k |C_k|^2)}. \quad (5)$$

There are 247 STFT frames per pool: 1,024-point periodic Hann, hop 256, per-frame mean subtraction, and FFT normalization by the window sum. No frame crosses a pool boundary.

Each target bin must have pooled energy at least 10^{-6} of the total non-DC coefficient energy. The recording scalar is the median of eligible pool values, or null if neither pool is eligible. Overlapping STFT frames are not independent observations.

For the frozen Native30 transfer view, decoded stereo FLOAT samples are promoted to float64 and arithmetically averaged. The entire 30 seconds is resampled from 44,100 to 16,000 Hz using a 160/441 polyphase ratio, Kaiser parameter 5 and constant-zero padding. Only then is the centre eight seconds selected, samples [176000, 304000) of the 480,000-sample view, followed by fixed gain 0.25. The two pools therefore cover seconds [11, 15) and [15, 19) of the Native30 input. There is no global DC subtraction, channel selection, normalization or fallback for cancellation. This view intentionally differs from the F/H mono loader. Full 228-cell arrays support replay, but the fixed target is the sole added predictor.

High bicoherence need not imply nonlinear generation: stable linear tones can also yield high values, and this statistic is amplitude-weighted rather than phase-only. A controlled external gate therefore motivates a fixed measurement transfer, not a claim about an AI generation mechanism.

7.5 Implementation references

All paths below are relative to `artifacts/audio_phenomena_expansion_20260907/` in the retained project. F: `code/phase_features.py`. H/M: `code/musical_features.py`. Stereo measurements: `code/stereo_candidate_v1.py`; the Native30 catalogue and joins are bound by `code/prepare_native30_evaluation_inputs_v3.py`. BC: `code/bicoherence_audio_v1.py`, `code/bicoherence_primitive_v2.py`, `code/bicoherence_scalar_v1.py` and `code/run_native30_bc_v2.py`. The prospective transfer specification is `BC_NATIVE30_INTEGRATION_DRAFT_EN.md`; its historical conditional language is distinct from subsequent measurement authorization and results. Scientific thresholds and feature definitions are not changed by this editorial integration.

8 External measurement validation and retained failures

The following assays address whether a measurement detects its proposed phenomenon. They do not directly estimate AI/Human classification accuracy. Passing software or integrity checks and satisfying a scientific gate are reported separately. Failed gates remain failed even when a subset of metrics or a subsequent exploratory classifier appears promising.

An early, small Music Flamingo pilot on MIDI-grounded drum comparisons failed the project's admission checks for dynamics and rhythm ranking. The model was therefore not included in the quantitative detector, and this limited trial does not support a broader assessment of its music understanding capabilities. Detailed results and logs remain in the standalone report¹; the thesis focuses on the subsequent measurement workflows.

8.1 Dynamics, rhythm change, and section variety

The controlled-change experiment tested variable gain, variable tempo, and variable section lengths, retaining constant-gain and constant-tempo negative controls. Its frozen family gate required a probability-of-superiority point estimate of at least 0.75 and a bootstrap 95% lower bound above 0.60.

Dynamics therefore has evidence of sensitivity to amplitude variation, not recovery of original MIDI velocities from mastered mixtures. Within R, tempo entropy responded individually (superiority 0.917 on 36 observable pairs), but that does not override the failed family gate. P

¹`artifacts/music_flamingo_midi_rhythm_probe_20260903/reports/FINAL_REPORT_EN.md`

Table 5: External controlled-change family results. Superiority concerns paired changes, not authorship classification.

Family	Pairs	Superiority	95% interval	Gate
Dynamics	50	0.880	[0.780, 0.960]	Pass
Rhythm change	44	0.705	[0.580, 0.818]	Fail
Section/bar variety	44	0.523	[0.466, 0.591]	Fail

was severely limited by observability: some section endpoints were available on only one positive pair. Reporting perfect ordering on that one pair would be misleading.

All three workflows were subsequently applied exploratorily to the existing 2,000-recording Human/Suno/HeartMuLa/ACE-Step collection. Pooled AUC was 0.875 for the spectral baseline, 0.892 with dynamics, and 0.914 with all three additions. These historical predictive results do not retrospectively qualify R or P as validated phenomenon measurements and are not Native30 results.

8.2 SC: low-band spatial response and codec controls

The low-band SC engineering gate used seven MedleyDB recordings held out from the earlier 24-recording spatial-control development set. The admitted candidate consists only of median absolute interchannel level difference and median side-energy fraction in three bands below 6 kHz. The analysis uses 44.1 kHz, a 4,096-sample periodic Hann window, hop 1,024, and no centring or padding. Higher-band, coherence, phase-increment and IQR diagnostics are not additional admitted predictors.

All 196 primary arithmetic checks and 28 eligible codec comparisons passed. Across six descriptors and 28 codec/recording pairs, 166 of 168 nuisance-to-target ratios were at most 0.10; the largest was 0.138683. Each descriptor met the frozen requirement that at least 90% of its ratios were at most 0.10 and none exceeded one. Independent replay recomputed the checks and ratios, verified 435 raw and three gate products, and checked 56 target-waveform hashes without importing the producer.

This threshold was selected after the development analysis, then frozen before these seven recordings. Development had shown substantial AAC128 sensitivity in the 6–12 kHz side-energy descriptor: median codec-to-width target ratio 0.621720, with six of 24 comparisons above one. The subsequent low-band restriction is therefore prospective relative to the seven-record gate, not a completely development-free choice. The seven tracks are artist-proxy/track-disjoint from the preceding controls but reuse MedleyDB; they are not a new independent AI/Human attribution population.

The result supports response to the specified spatial interventions relative to the tested codecs. It does not establish universal codec invariance, panning-motion understanding or authorship detection. The later Equal60 matched SC increments provide the separate predictive test and retain negative transfer effects. Evidence under

artifacts/audio_phenomena_expansion_20260907/:
 STEREO_LOWBAND_GATE_V2_RESULTS_EN.md,
 preregistration/stereo_lowband_gate_frozen_v2.json,
 results/stereo_lowband_gate_v2/gate_result.json,
 code/audit_stereo_lowband_gate_v2.py, and
 audit/stereo_lowband_gate_parent_replay_v2.json.

8.3 Breath events: recall without usable precision

The retained breath benchmark used 113 reviewed clips, twenty singer-prefix groups, 141 merged high/medium reference events, and 38 clips with no reference event. A known archive-identity

conflict was excluded before this evaluation; the conflicting candidates and exclusion evidence were retained.

Independent maximum-bipartite event matching required predicted duration at most two seconds, onset error at most 200 ms, and positive temporal overlap. It reproduced 137 true positives, 1,257 false positives, and four false negatives. Precision was 0.098278, recall 0.971631, and F1 0.178502. Both precision and F1 were below their required 0.70 threshold. All 38 no-event clips had at least one detection, contributing 352 false-positive events.

Aggregate frame BA of 0.912017 was a diagnostic on a different unit and could not substitute for event precision or F1. The integrity/recomputation audit passed, but the measurement gate failed. B was not promoted to AI/Human classification on this evidence.

8.4 Vibrato: differential observability

The VocalSet development assay opened 43 pairs (86 isolated monophonic recordings from ten singers), while 47 evaluation pairs remained unopened. Pairing used singer and repertoire/exercise/vowel identity, not time-aligned or guaranteed equal-pitch performances. The subset had originally been selected for breath annotation and was not a random sample of all VocalSet.

The frozen coarse pYIN workflow used 16-kHz audio, 1,024-sample frames, 160-sample hops, and a C2–C6 range. It produced a descriptor for 42/43 straight recordings (97.67%) but only 37/43 vibrato recordings (86.05%). The 11.63 percentage-point availability gap exceeded the predeclared ten-point limit. Among the 37 jointly observable pairs, differences were positive in 33, tied in three, and negative in one. This favorable conditional ordering could not bypass the observability failure. An independent reconciliation passed 883 checks; it did not establish continuous-F0 ground-truth fidelity. V remained unadmitted, and the reserved pairs were not used to rescue the candidate.

8.5 Envelope modulation: a near-threshold result remains a failure

The reserved Q assay used 52 distinct-PCM notes from 26 instrument identities, eight fixed interventions, and three views: original, MP3 128 kbit/s, and AAC-LC requested at 64 kbit/s, yielding 1,248 measurements. These were reused NSynth sample-library notes, not an independent performed-song domain.

The six descriptors measured fast modulation-power fraction and modulation entropy in three carrier bands: 250–1,000, 1,000–3,000, and 3,000–7,000 Hz. Carrier frequency and modulation frequency are different quantities. The entropy intervention compared chirped amplitude modulation with 48-Hz amplitude modulation at depth 0.6; the required note-level increase was 0.10, with at least 90% instrument-balanced success.

Of 146 prospective checks, 141 passed and five failed. High-carrier entropy success was 88.462% in all three views, below 90%. Two additional AAC nuisance checks failed for low- and high-carrier entropy. The maximum available nuisance ratios remained below one; the failures concerned the required fraction at or below 0.1, not catastrophic ratios above one. Passing the remaining rate, depth, gain, polarity, and eligibility checks did not qualify the full Q family. Selecting only successful descriptors after inspection would not inherit the original gate’s validity.

The consistency audit checked 6,659 products and recomputed all 146 decisions. A separate DSP replay reproduced 1,875,744 spectrum bins with zero numerical discrepancy using the same NumPy/SciPy libraries. This was not an independent-library replication. Requested codec bitrates were not assumed equal to observed stream rates, and the two recipes were not a bitrate-matched codec ranking.

8.6 Bicoherence: a passed controlled measurement gate

BC measures consistent spectral coupling at the fixed 500/750/1,250-Hz triplet. The reserved GuitarSet assay used 90 recordings and completed 1,440 measurements without failed or skipped cases: 630 float64, 270 float32 precision controls, and 540 codec-decoded measurements. All 72 scientific checks were satisfied after independent numerical/accountability replay.

The closed-coupled minus independent-phase contrast was 0.863033 at -6 dB and 0.936507 at 0 dB; all ninety recording contrasts were positive. Baseline median absolute scalar errors were 0.000299 for MP3 128k and 0.000634 for Opus 96k, with respective 95th percentiles 0.001563 and 0.005402. Eligibility mask agreement was 1.0 under these baseline codec comparisons. These numbers are controlled measurement responses and errors, not classification AUC.

An initial independent audit failed on a raw complex-sum mantissa comparison. The corrected auditor made normalized column storage explicitly F-order and asserted contiguity, preserving scientific tolerances, thresholds and producer products. The original failure, diagnosis, correction authority and successful replay were retained. Audit success and scientific gate satisfaction were not treated as automatic classifier admission: Native30 transfer required a separate fixed-view measurement protocol.

The controlled guitar backgrounds do not establish universal validity on mastered mixtures. Arithmetic stereo downmix can cancel content, bicoherence is nonlinear, and phase-stable linear tones can produce a high statistic. The result supports a limited fixed-statistic measurement capability, not proof of a generative mechanism or established AI/Human utility.

8.7 References to retained experiments

The D/R/P protocol, paired measurements, and application results are in `artifacts/dynamics_rhythm_change_detector_extension_20260903/`, including `reports/REPORT_EN.md` and the retained controlled-change CSVs. The remaining reports are under `artifacts/audio_phenomena_expansion_20260907/`: `BREATH_BENCHMARK_RESULTS_EN.md`, `VOCALSET_VIBRATO_DEVELOPMENT_RESULTS_EN.md`, `results/modulation_nsynth_gate_report_v1/RESULTS_EN.md`, and `results/bc_reserved_verified_report_v2/REPORT_EN.md`. These summaries point to their frozen implementations, protocols, measurements and independent audits; they do not replace the underlying evidence.

9 Completed development experiments and transfer boundaries

This section reports the historical Exact10 and Equal60 experiments, separately from the later Native30 evaluations. The exact-ten-second and exact-sixty-second cohorts differ; their scores are not a controlled comparison of listening duration. All results retain their original development status. In particular, inspecting many feature subsets does not create an untouched validation set for the selected maximum.

9.1 Exact10: conditional tonal-path gains

The five-family S/D/R/F/H experiment evaluated 31 nonempty subsets at five training caps on 8,361 development recordings: 2,161 Human recordings from seven sources and 6,200 AI recordings from thirteen sources. P was unavailable at ten seconds and was not included. At the all-data cap, S+D had $J = 0.632004$ and mean BA = 0.596233; S+D+H reached $J = 0.660124$ and mean BA = 0.616698. These are the finalized Exact10 values, distinct from the slightly different historical four-family evaluation.

The analysis used 1,000 paired global-group draws and 10,000 separate crossed-source draws, retaining the fitted models and threshold. Because 500 groups crossed labels, the union contained

Table 6: Matched additions to S+D in Exact10. The group intervals condition on fixed fitted models. Source ranges separately describe observed-source composition sensitivity, not confidence intervals for future generators.

Addition	ΔJ	Group 95% interval	Source 95% range
F	-0.007614	$[-0.019946, 0.003516]$	$[-0.060936, 0.033818]$
H	+0.028120	$[0.018631, 0.036721]$	$[-0.006843, 0.063114]$
F+H	+0.024647	$[0.010972, 0.038292]$	$[-0.022888, 0.069376]$

2,339 groups, not the sum of the within-class group counts. Four group draws generated 104 explicitly undefined pair cells; none was replaced by a chance score. All aggregate differences remained finite under the declared aggregation policy.

The H and F+H conditional group intervals were positive, but every source composition range crossed zero (Table 6). Thus the evidence supports a conditional improvement on these observations, not stable improvement under arbitrary source shift. The uncertainty protocol was frozen after point estimates were visible and is not a multiplicity-adjusted discovery claim. The classifier audit passed 21,380 checks; a separate saved-output verification passed 78,111 assertions. These audits establish specified numerical and accounting properties, not universal detection validity.

9.2 Equal60 v6: stereo is not uniformly additive

The native-stereo, exact-sixty-second cohort contained 2,174 recordings: 1,278 Human and 896 AI. Human sources were MTG-Jamendo (481), MAESTRO (300), MedleyDB (156), MoisesDB (238), and Saraga (103). AI sources were Suno (396) and Mureka (500). Mureka and Saraga had already participated in exploratory development. Each generator-held-out arm trained on the other generator; this is not an external test on a third generator.

Eight families supplied 60 descriptors: S (15), D (3), R (3), P (6), F (15), H (6), M (6), and SC (6). Neither BC nor failed modulation family Q was included. All 255 nonempty subsets and five caps were evaluated under 38 valid folds: 23 Human-source, ten generator, and five ordinary-group folds. The two empty Saraga fold cells remained omitted. Comparisons below are within v6; they do not subtract scores from the different v5 population.

Table 7: Equal60 v6 all-cap results. Cells give AUC / BA (%). Seven is S+D+R+P+F+H+M; Eight adds SC. Directions are different evaluation estimands.

Set	Human-source	Generator	Ordinary-group
S	0.7882 / 72.07	0.6416 / 52.56	0.8708 / 81.20
D	0.7206 / 67.10	0.6339 / 61.48	0.7334 / 68.65
R	0.7445 / 67.66	0.7331 / 67.37	0.7330 / 67.33
P	0.6907 / 64.41	0.6408 / 60.68	0.6891 / 64.41
F	0.8304 / 75.72	0.6769 / 57.75	0.8580 / 77.94
H	0.7343 / 67.94	0.7076 / 66.38	0.7686 / 71.39
M	0.5153 / 52.42	0.5624 / 54.40	0.6467 / 60.15
SC	0.7173 / 72.37	0.7230 / 71.88	0.7784 / 77.43
S+D+R+P	0.8545 / 76.72	0.7532 / 58.80	0.9178 / 85.64
Seven	0.9071 / 81.17	0.7736 / 57.32	0.9482 / 88.67
Eight	0.9056 / 81.97	0.7708 / 56.51	0.9506 / 89.66

SC alone had generator-held-out BA of 71.88%, yet adding it to Seven decreased that end-point from 57.32% to 56.51%. Ordinary-group BA increased from 88.67% to 89.66%. A stand-alone association is therefore not sufficient to predict a beneficial multivariate increment. Across

all 127 matched SC additions at the all cap, generator BA increased for 64 subsets and decreased for 63. These correlated descriptive counts are not independent trials or significance tests.

Table 8: Matched Equal60 v6 generator-held-out BA by training cap. Adding SC to Seven reduced this endpoint at every cap; more groups were not monotonically beneficial. Caps are global groups per source.

Cap	Seven BA (%)	Eight BA (%)
25	57.98	57.22
50	61.17	59.87
100	57.28	55.57
200	58.54	57.50
All	57.32	56.51

9.3 Errors concealed by an ordinary-group summary

Under Eight, held-out Mureka sensitivity was 136/500 (27.20%) and held-out Suno sensitivity was 63/396 (15.91%). Human specificity was 41/103 (39.81%) on Saraga and 273/481 (56.76%) on MTG-Jamendo. These rates belong to separate fitted held-source arms and cannot be combined into one confusion matrix. They demonstrate why the 89.66% ordinary-group BA cannot serve as a claim of reliable unseen-generator attribution.

SC missingness-only AUC and BA were 0.5 in all three directions, while its values-only results equalled its primary results. This excludes one specific missingness explanation for SC within this cohort, but not genre, panning, mastering, codec, or source effects. Eight’s generator-held-out BA was 56.26% in the values-only diagnostic and 56.32% in the missingness-only diagnostic, versus 56.51% in the primary representation. These are separately fitted diagnostics, not an additive decomposition of the primary score.

The Equal60 numerical audit covered 67,830 models, 19,624,290 score rows and 60,690 OOF endpoint cells. Maximum affine-score replay discrepancy was 3.997×10^{-15} . Checks of training transformations, identities, denominators, and normal-equation residuals support numerical consistency; stationarity checks do not independently reproduce an optimizer or establish causal authorship attribution.

9.4 Result and implementation references

All paths in this paragraph are relative to the retained experiment root `artifacts/audio_phenomena_expansion_20260907/`. Exact10 full tables are in `results/presentation_sdrfh_10s_v2/`; conditional and source uncertainty outputs are in `results/uncertainty_sdrfh10_v1/`, implemented by `code/bootstrap_sdrfh10_uncertainty.py`. Equal60 saved models and scores are in `results/equal60_v6_classification_v1/`, with numerical audit `audit/equal60_v6_classification_v1.json`. The complete 255-subset tables and matched increments are retained in `results/presentation_equal60_v6_v1/`, produced by `code/present_equal60_v6_results.py`. These references distinguish measurement outputs, analysis outputs and code; the final appendix supplies the full artifact and hash index.

10 Native30 seven-family results

This section reports the completed seven-family Native30 evaluation, not the subsequent BC or YuE2 extension. The cohort contains 3,830 recordings: 1,664 human and 2,166 AI, drawn from eleven sources. The committed evaluation contains 51,562 valid model instances (36,830 primary

and 14,732 diagnostic), with zero recorded failed model instances. There are 127 nonempty feature subsets, five training caps, and separately defined source-held-out and grouped-descriptive protocols.

10.1 Endpoint and reporting convention

The primary mode combines descriptor values with missingness indicators. Training rows alone determine medians and weighted scaling. Ridge regularization is fixed at 10 and the decision threshold at 0.5. Scores are unbounded decision values, not calibrated probabilities. No held-source threshold tuning or post-hoc winner selection is performed.

Tables report balanced accuracy (BA) in percent. Correctness is averaged within global groups, then equally across sources within each class, and finally equally across classes. Protocol summaries weight held-source experiments equally. The caps constrain unique global groups per remaining training source, not total recordings. Transitive source and test dependencies are excluded from training. The exhaustive 1,905 primary cells remain available; the tables below show the singleton families, the original four-family set, and the full seven-family set without claiming that these are optimal.

S denotes common-band vocal spectral descriptors; D onset-conditioned dynamics; R rhythm; P phrase/section-length descriptors; F phase/group delay; H pitch-class/chroma organization; and SC stereo descriptors. H does not measure acoustic overtone shape.

10.2 Generator holdout

Features	25	50	100	200	All
S	61.15	62.02	62.18	61.56	61.39
D	58.36	58.86	59.04	58.60	59.02
R	61.30	61.23	61.63	62.04	62.00
P	53.28	52.27	51.70	52.92	52.84
F	58.71	57.28	56.43	56.81	56.72
H	52.77	51.13	50.84	50.05	50.42
SC	52.19	53.12	52.96	52.22	52.20
S+D+R+P	65.41	64.96	66.02	65.87	65.79
S+D+R+P+F+H+SC	61.87	59.71	59.77	61.27	61.63

10.3 Human source holdout

Features	25	50	100	200	All
S	59.72	60.62	60.63	61.09	60.95
D	62.20	62.14	62.21	62.40	62.40
R	61.12	61.21	61.22	61.55	61.44
P	54.38	54.64	54.57	54.55	54.48
F	61.33	60.15	59.77	59.42	59.27
H	56.91	58.11	57.20	57.59	57.51
SC	58.04	60.35	60.94	60.39	60.28
S+D+R+P	63.63	64.94	66.50	66.03	65.65
S+D+R+P+F+H+SC	65.57	66.43	67.58	68.02	67.99

10.4 Ordinary group holdout descriptive

Features	25	50	100	200	All
S	69.36	70.48	69.83	70.00	70.26
D	63.34	63.19	63.17	62.91	63.04
R	62.12	62.01	62.02	62.53	62.61
P	54.40	54.83	54.88	54.94	54.90
F	67.22	66.47	66.68	67.33	67.09
H	60.31	60.41	59.95	58.77	58.86
SC	62.10	62.77	63.13	62.63	63.01
S+D+R+P	72.09	74.03	75.25	74.59	74.30
S+D+R+P+F+H+SC	74.84	75.93	75.76	75.95	75.66

10.5 Interpretation

At the all-data cap, the four-family model obtains 65.79% BA under generator holdout, 65.65% under human-source holdout, and 74.30% under ordinary grouped descriptive testing. The full seven-family model obtains 61.63%, 67.99%, and 75.66%, respectively. Thus the additional families improve the displayed human-source and grouped summaries while reducing the displayed held-generator endpoint. More interpretable descriptors are not automatically more transferable descriptors.

The differences are descriptive comparisons within the recorded protocol, not causal effects of adding a phenomenon to music. No confidence interval is inferred from five fixed buckets. AUC is retained within each fitted fold; raw scores from different models are not pooled into one global AUC. Repeated opposite-class predictions across held-source experiments are dependent occurrences, not additional independent recordings.

The training-cap curves are not monotonic. Selecting the largest observed cell after inspecting the full lattice would require separate untouched confirmation before an optimality claim. These results neither establish a universal AI fingerprint nor justify an accusation against an individual musician. Scientific validation of the intended dynamics, rhythm, and phrase measurements remains distinct from their classification usefulness.

10.6 Reproduction references and completed extensions

Code: `artifacts/audio_phenomena_expansion_20260907/code/evaluate_native30_v3.py` and `summarize_native30_evaluation_v3.py` in the same directory. Full report outputs: `artifacts/audio_phenomena_expansion_20260907/native30_evaluation_report_v3_20260911/`. The source `summary.json` and `COMMIT.json` bind the reported cells. The compact CSV is `deliverables/overnight_snapshot_20260912_v1/all_seven_family_primary_cells.csv`.

The public archive is [the EZMONYI HF evaluation dataset](#). Intermediate reports remain preserved alongside later completed results; full-corpus audio publication is still incomplete. The completed eight-family BC comparison, YuE2 transfer, and expanded-cohort evaluations are reported in the following sections, with their own terminal result bindings. Metadata schedules alone are not evidence of completed fits.

11 Completed Native30 bicoherence extension

The completed extension evaluates all 255 nonempty subsets of eight families (S, D, R, P, F, H, SC and BC), with 55 descriptors on the original 3,830-recording Native30 cohort. BC is the fixed center-eight-second median squared bicoherence statistic at the predeclared 500/750/1,250-Hz coupling coordinates; it is not a scan selecting the most discriminative frequency triplet. All five training-group caps and the three distinct grouped evaluation protocols are retained. There are

103,530 completed model evaluations: 73,950 primary and 29,580 diagnostic cells. The 3,570 structurally ineligible cells are recorded as omissions, not silently treated as completed trials.

Each fitted fold uses training-derived imputation and scaling, ridge parameter 10, a fixed threshold of 0.5 and uncalibrated linear scores. Primary balanced accuracy averages correctness within groups, then equally across sources within each class, then across the two classes; held-source experiments are equally weighted. AUC is calculated within individual fitted folds, never by pooling scores from different models. Repeated opposite-class test occurrences do not constitute additional independent recordings.

Feature set	Generator holdout	Human-source holdout	Grouped descriptive
S	61.39	60.95	70.26
D	59.02	62.40	63.04
R	62.00	61.44	62.61
P	52.84	54.48	54.90
F	56.72	59.27	67.09
H	50.42	57.51	58.86
SC	52.20	60.28	63.01
BC	44.62	45.96	49.62
S+D+R+P	65.79	65.65	74.30
S+D+R+P+BC	65.05	65.19	74.20
S+D+R+P+F+H+SC	61.63	67.99	75.66
S+D+R+P+F+H+SC+BC	61.40	67.65	75.80

Table 9: Native30 group/source-balanced accuracy (%) at the all-cap configuration. Protocols have distinct estimands; no cross-column causal comparison is implied.

Protocol (all eight families)	25	50	100	200	All
Generator holdout	62.47	59.30	59.41	61.02	61.40
Human-source holdout	65.43	66.52	67.07	67.69	67.65
Grouped descriptive	74.77	76.43	75.95	75.78	75.80

Table 10: All-eight-family training-group-cap ablation. Caps are per-source group budgets, not total training-song counts. The complete 3,825-cell table retains all 255 subsets at all five caps.

Adding BC to the seven-family configuration changes generator-held-out balanced accuracy from 61.6251% to 61.3984% (-0.2267 percentage points), and human-source-held-out balanced accuracy from 67.9920% to 67.6457% (-0.3462 points). Ordinary grouped descriptive performance changes from 75.6649% to 75.8033% ($+0.1384$ points). BC alone reaches 44.6224%, 45.9584% and 49.6200%, respectively. These are descriptive outcomes at the fixed operating point, not significance tests. Below-chance performance is reported as observed; labels and score signs are not reversed after testing. The small grouped-descriptive increase does not establish better transfer.

The report preserves every subset and cap rather than naming a best classifier. The completed matrix provides no evidence that BC consistently improves cross-source discrimination. External phase-coupling sensitivity and robust AI/human attribution remain different claims. In particular, a measurable coupling statistic need not identify generative provenance.

Admission terminology and evidence limits. The legacy reporting schema retains the flag `BC_classifier_admitted=false`. This must not be read as saying BC was never supplied to an exploratory fitted model: BC-containing subsets were explicitly evaluated. It means that the experiment does not establish BC as a validated general-purpose detector. Report generation performed no fitting, model selection or threshold tuning. Product hashes and all evaluation

receipts were checked, but the report does not claim a separately implemented numerical audit of every aggregate metric or confidence intervals over fixed buckets.

Reproducibility. The full report is retained locally under `current_results/native30_bc_evaluation_report_v3_20260912/`; its COMMIT SHA256 is:

```
ad5223a5f1f0fb667fe38a1e0c214b28ab8a8d0b57f01f912521885aa1c0213b
```

The underlying evaluation COMMIT is:

```
6d6acecbb11b6fcee72f7ba8730b1a2a26f2797c38d659ca69151074dd3dba06
```

The compact CSV in `current_results/bc_thesis_tables_v1/` retains all 3,825 primary protocol/subset/cap cells, including denominators and omissions. The exporter is `export_bc_thesis_tables_v1.py`; the source reporter is `summarize_native30_evaluation_bc_v3.py`. Earlier failed report adapters and earlier thesis versions remain preserved rather than overwritten.

12 Expanded Native30 evaluation including YuE2

The completed expanded development cohort contains 4,228 recordings: the original 3,830 plus 398 eligible YuE2 development recordings. Another 100 YuE2 recordings remain a separate locked population, evaluated in the prediction-only sensitivity analysis rather than inserted into this training matrix. Two short originals are excluded from Native30. Shared Muse source-song groups link generated recordings across models and are retained in the group/component split restrictions.

All 255 subsets of S, D, R, P, F, H, SC and BC are evaluated under five training-group caps and the three protocols. The completed evaluation comprises 112,455 model instances (80,325 primary; 32,130 diagnostic), with 3,570 structurally ineligible scheduled cells explicitly omitted. The reporting projection contains 3,825 primary protocol/subset/cap cells. Training-only transforms, fixed ridge parameter 10, threshold 0.5, null-value handling and the group/source-balanced estimand follow the preceding Native30 protocol.

Feature set	Generator holdout	Human-source holdout	Grouped descriptive
S	65.83	64.34	71.98
D	59.23	63.42	63.58
R	62.98	62.53	63.20
P	53.32	54.42	54.78
F	58.94	62.11	68.67
H	54.69	59.51	62.40
SC	56.07	62.63	64.76
BC	46.11	45.59	48.02
S+D+R+P	69.72	68.63	76.37
S+D+R+P+BC	69.10	68.26	76.19
S+D+R+P+F+H+SC	66.18	70.21	77.87
S+D+R+P+F+H+SC+BC	66.29	70.25	78.37

Table 11: Expanded Native30 group/source-balanced accuracy (%) at the all-cap configuration. Protocols have distinct estimands; no cross-column causal comparison is implied.

At the all-cap configuration, S/D/R/P attains 69.7234% balanced accuracy in generator holdout, compared with 66.1827% for seven families and 66.2877% for eight. In human-source holdout, the corresponding values are 68.6270%, 70.2078% and 70.2487%; ordinary grouped descriptive values are 76.3652%, 77.8734% and 78.3713%. Thus broader feature combinations

Protocol (all eight families)	25	50	100	200	All
Generator holdout	65.46	64.11	64.48	65.94	66.29
Human-source holdout	67.62	69.05	70.28	70.22	70.25
Grouped descriptive	76.66	78.22	77.90	78.30	78.37

Table 12: All-eight-family training-group-cap ablation. Caps are per-source group budgets, not total training-song counts. The complete 3,825-cell table retains all 255 subsets at all five caps.

have different effects under different generalization targets. The results do not justify a single universally superior configuration.

BC adds only 0.1050 percentage points to the seven-family generator-holdout result and 0.0408 points to human-source holdout, versus 0.4980 points in the ordinary grouped protocol. These small descriptive differences are not significance evidence or proof that BC is a transferable authorship signature. BC alone remains below 50% in all three protocols at the fixed operating point.

Comparison with the pre-YuE2 cohort. The generator-holdout S/D/R/P result changes from 65.7872% to 69.7234%. However, the expanded generator set changes both training composition and the equally weighted held-generator test population. This is not a fixed-test-set learning-curve estimate and cannot identify the causal benefit of adding 398 recordings. Similarly, altered component structure and missingness affect eligible folds. The within-cohort cap tables and complete subset tables are retained; choosing the best observed cap would require a further independent selection/evaluation design.

Artifacts and evidence limits. The full report is retained under `current_results/expanded_native30_report_v1/`. Its COMMIT SHA256 is:

b7934f2c649e5d0fbada832c335d36839a1aacb3bd7d6a674db83af5372bc911

All seven committed report products were verified after local transfer. The original model-evaluation COMMIT is:

11f074f867d0e3721c6a586e8c8f86116ff1b0198147ba14f6379b5d6f55570d

`export_yue2_thesis_tables_v1.py` generates the complete 3,825-row CSV and the displayed tables from that committed report, without refitting or selection. The CSV resides in `current_results/yue2_expanded_thesis_tables_v1/`. Reporting is not a separately implemented audit of all aggregate numerical calculations, and no confidence intervals or pooled cross-model AUC are claimed. Native60 transfer is a separate duration-selected experiment, reported in Section 15.

13 YuE2: prediction-only transfer to locked recordings

The extension generated 500 recordings from the shared frozen Muse prompts. Of these, 498 supported a native 30-second view: 398 development recordings and 100 locked recordings. Two shorter recordings were excluded from this Native30 analysis, not padded into eligibility. The experiment reported here uses the 100 locked AI recordings only and does not use the expanded models.

13.1 Fixed model catalogue and protocol deviation

All 255 nonempty subsets of the eight pre-existing descriptor families were evaluated with five previously fitted ordinary-group-holdout models per subset. These models use the original cohort, the all-quantity setting, values plus missingness, and a fixed decision threshold of 0.5.

Training medians, scaling and coefficients were reused, with zero new fits. The preflight checked disjoint item, group and component identities between training and the locked recordings. Each prediction was replayed deterministically. The result has 1,275 models and 127,500 prediction occurrences, but only 100 test recordings.

The originally proposed order was to score the held-out generator before expanded development. Actual locked scoring occurred afterwards. This is a protocol-order deviation, and the present report does not retrospectively claim prospective preregistration. The exhaustive scoring catalogue was fixed before this scoring run; no subset was chosen as a winner from these outcomes.

13.2 Sensitivity, not two-class accuracy

For a fixed model, sensitivity is the fraction of these 100 AI recordings predicted as AI. There are no human test recordings in this experiment, so specificity, balanced accuracy and ROC AUC are not estimable. A trivial always-AI predictor would also achieve perfect sensitivity. Table 13 reports the single families and two pre-existing combinations; the exhaustive 255-subset table is retained separately.

Table 13: Locked YuE2 sensitivity (percent). Means and ranges are descriptive across five old models evaluated on the same recordings, not confidence intervals or independent test-set repetitions.

Family/subset	Mean	Range
S	94.8	93–97
D	62.6	62–65
R	84.8	84–85
P	28.0	28–28
F	87.8	85–90
H	87.0	85–90
SC	90.2	88–91
BC	66.8	29–100
S+D+R+P	96.0	95–97
S+D+R+P+F+H+SC+BC	96.2	95–97

The four-family and eight-family combinations have similar descriptive sensitivity, but this does not establish equal overall performance or an incremental benefit from the additional families. BC alone varies strongly across the old models. It must be assessed jointly with human false-positive rates before interpreting utility. The results do not identify the acoustic cause of the label association and cannot establish an intrinsic AI signature.

13.3 Reproduction and retained evidence

The code repository is <https://github.com/ezmonyi/ai-music-detection>. The versioned scripts are `current/yue2_500_extension_20260911/preflight_locked_yue2_v1.py`, `score_locked_yue2_v1.py`, and `report_locked_yue2_v1.py` in that same directory. The scoring protocol and freeze are retained alongside them. The source scoring COMMIT SHA256 is `b43744c0775c14d040579537316b40d1c6a3479649afc285faa176e76abf87f5`. All 1,276 committed products were verified after local recovery.

Under `/Users/yi/Documents/report/music_ai_detection_20260912/`, `current_results/locked_yue2_scoring_v1/` contains individual predictions and `current_results/locked_yue2_report_v1/all_255_combinations.csv` contains the complete descriptive

comparison. These paths identify preserved results, not a claim that all remaining experiment arms are finished.

14 YuE2 legacy spectral replication and separator correction

This separate replication uses the historical first-30-second, 44.1 kHz stereo representation, including padding for two short YuE2 originals. It is not interchangeable with the native-center30 cohort. Five hundred human and 500 YuE2 recordings contribute full-mix descriptors and raw/corrected vocal descriptors; the separator calibration is reused from external MUSDB oracle vocals, not estimated from this AI/human comparison.

14.1 Fixed within-generator comparison

The fixed development/test split has 400/100 recordings per class. Human artist groups and YuE2 source-song groups are disjoint across the split, but the human test recordings have been used in historical experiments. Moreover, YuE2 appears in development. Thus this is a within-generator historical-holdout comparison, not a new external or held-generator validation.

The original legacy ridge implementation fits targets $2y-1$ with penalty $\alpha = 10$ and threshold zero, using training-only means and scales. An initial adapter incorrectly assumed the later Native30 threshold of 0.5; an independent balanced-accuracy calculation stopped that run. The preserved v2 adapter restores the original zero threshold; it does not optimize a threshold using test outcomes. The original row-wise development cross-validation is not presented here as group-independent evidence.

Table 14: The fixed 22-statistic spectral representations on 100 human and 100 YuE2 test recordings. All 18 pre-existing full-mix/vocal and frequency-band representations are retained in the companion CSV; these three rows are a matched representation comparison, not the top-ranked outcomes.

Representation	BA	AUC	Specificity	Sensitivity
Full mix	0.925	0.9706	0.940	0.910
Raw vocal	0.950	0.9837	0.930	0.970
Corrected vocal	0.950	0.9832	0.930	0.970

In this matched comparison, separator correction does not change balanced accuracy and changes AUC only slightly. These point estimates do not establish statistical equivalence, causal attribution, or benefit on unseen generators. In particular, source composition, production, mastering, vocal activity and excerpt position can contribute to the observed high discrimination.

14.2 Average difference spectrogram

Figure 1 compares class-average dB spectra on the locked subset. The STFT uses 4096 samples and a 1024-sample hop; displayed frequencies are aggregated into 128 logarithmic bands from 300 Hz to 20 kHz. Shape panels subtract each frame’s mean across these bands. Clips are aligned by excerpt time, not by musical phrase or vocal onset. The marked time dependence therefore requires caution: the plot does not isolate a stationary generator fingerprint. Subtracting a shared frequency response from both classes also need not remove their class contrast; the correction is not a learned AI-minus-human template.

14.3 Retained code and outputs

Under the report root `/Users/yi/Documents/report/music_ai_detection_20260912/current_results/`, `legacy_spectral_measurement_v1/` retains the metrics, frequency

vectors, three cohort heatmaps, numerical maps and input bindings; `legacy_fullmix_v1/` retains the full-mix control; `legacy_spectral_locked_v2/` retains all fitted parameters and predictions; `legacy_spectral_report_v2/all_representations.csv` gives all 18 results. The failed first adapter is retained in `legacy_spectral_locked_v1_failed/`. Local product hashes were verified against their producing COMMIT records before this chapter was written. Relevant source scripts are `run_legacy_spectral_measurement_v1.py`, `extract_legacy_fullmix_v1.py`, `evaluate_legacy_spectral_locked_v2.py`, and `report_legacy_spectral_locked_v2.py`, in the YuE2 extension directory.

15 Completed frozen-model transfer on YuE2 Native60

Of the 500 original YuE2 outputs, 276 support an exact native center-60-second interval: 223 development and 53 locked recordings. The remaining 224 are excluded by duration, not padded. Every crop was checked against the original 48-kHz stereo audio. Neural front ends use explicitly resampled 44.1-kHz views; F/H/M extraction uses the historical mono 16-kHz analysis loader. These are different documented views of the same interval, not interchangeable files.

All 276 neural and SDRP outputs were completed and hash-verified. Repeating the original SDRP extractor produced exactly matching descriptor values on all records; F/H/M values were reconciled with their committed per-record outputs. This checks reproducibility and record consistency, not a separately implemented acoustic algorithm. F/H/M audio extraction was not repeated in that final audit.

The historical equal60 model publication was independently re-audited; its 1,604-recording development population has no ID or group overlap with these YuE2 records. The transfer applies 127 subsets at five original caps and five stored folds (3,175 model applications), without fitting or threshold changes. The historical families are S/D/R/P/F/H/M, not the newer SC/BC extension. The 63 non-M subsets are reported separately from 64 M-containing subsets, which remain secondary historical diagnostics. M is not thereby admitted as a validated general detector family.

All 876,300 per-record predictions were checked against a second, scalar implementation of the saved imputation, missing indicators, scaling and linear coefficients. Maximum score discrepancy was 2.22×10^{-15} and all threshold decisions agreed. Persisted predictions were then independently counted and matched to all 9,525 per-model/population summaries. Repetition across models does not create 876,300 independent audio samples.

Feature set	Locked (53)	Development (223)	All eligible (276)
S	98.11	97.22	97.39
D	43.77	40.09	40.80
R	83.40	80.27	80.87
P	81.51	83.32	82.97
F	44.91	48.79	48.04
H	23.77	26.55	26.01
M	16.23	18.39	17.97
S+D+R+P	91.70	92.02	91.96
S+D+R+P+F+H	97.36	95.78	96.09
S+D+R+P+F+H+M	92.45	94.62	94.20

Table 15: YuE2 Native60 mean AI sensitivity (%) over five frozen models at the all-cap setting. These are single-class results, not balanced accuracy. M-containing rows are secondary historical diagnostics.

The locked-population all-cap mean sensitivity is 91.6981% for S/D/R/P, 97.3585% for S/D/R/P/F/H and 92.4528% for S/D/R/P/F/H/M. S alone reaches 98.1132%. These ob-

servations do not establish a superior binary classifier: human specificity, balanced accuracy and ROC AUC are not estimable from an AI-only population. The corresponding all-eligible sensitivities are 91.9565%, 96.0870%, 94.2029% and 97.3913%. All caps and subsets, including low single-family sensitivities, remain available in the full table; no winner is selected from this transfer set.

Length eligibility selects only 55.2% of the generated corpus and can correlate with musical structure. The 53 locked records are a subset of the earlier 100-recording locked population, not a newly acquired independent test set. Earlier Native30 and legacy analyses preceded this transfer. Differences from Native30 cannot be attributed solely to duration, because population selection, historical model training and feature definitions also differ.

Preserved outputs. `current_results/native60_transfer_scores_v1/` stores compressed individual predictions and complete model summaries. Its COMMIT SHA256 is:

```
34782ccfa467d5304ec2a4af1000d390eb467f004bb591ddce1474d06a068e64
```

`current_results/native60_transfer_report_v1/` stores 1,905 aggregate cells (127 subsets, five caps, three populations). The scripts are `score_native60_transfer_v1.py`, `native60_score_kernel_v1.py` and `report_native60_transfer_v1.py`. Input preparation, neural extraction, measurement replay and model-audit scripts and receipts are retained separately.

16 Synthesis and remaining limitations

The completed experiments show that interpretable measurements can provide label-associated information without uniquely identifying authorship. The Native30 results reinforce a central generalization limit: increasing feature count or training quantity does not guarantee better held-generator decisions. External measurement validation, source-held-out classification, and faithful reproduction are separate requirements. No descriptor family should be accepted as a causal AI signature merely because it improves one fitted classifier.

Native-channel and duration eligibility select particular populations. Source labels incompletely represent genre, language, culture, instrumentation, editing, codec, and mastering practice. Pretrained separators and music-understanding front ends add their own domain sensitivity. Human-produced music may use MIDI, synthesis, and extensive processing. These detectors should not be used as sole evidence to penalize a musician or certify a recording.

The completed Native60 transfer adds an AI-only, duration-selected endpoint; it does not resolve the human false-positive question. The YuE2 extension has 398 eligible development recordings and 100 separately retained locked-test recordings; two shorter originals are excluded from Native30 rather than padded. Shared prompt dependencies must remain excluded across training and evaluation.

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A Reproducibility and artifact navigation

The project retains executable definitions, experiment-specific manifests, raw responses or measurements, classifier outputs, audits and rendered reports. The canonical local workspace is `/Users/yi/Documents/code/music`. Paths beginning with `artifacts/` are relative to that root. Large audio and intermediate products can reside on the remote NFS data volume rather than the local report mirror. A local report’s existence does not establish that its full remote audio dependency graph is locally available.

A.1 Data identities, views and storage

The September 5 registry summary is `artifacts/source_diversity_expansion_20260905/results/inventory/corpus_inventory.json`, with readable companion `corpus_inventory.md` in the same directory. It records a particular frozen population; it is not a live total formed by adding later overlapping cohorts. The local inventory snapshot is `artifacts/audio_phenomena_expansion_20260907/audit/closeout_local_inventory_20260908_v1/files.jsonl`. It lists files, not independent songs or a complete remote mirror. Remote expansion code and logs use `/mnt/nfs-code/users/yi/audio_phenomena_expansion_20260907`; its primary audio/measurement products use `/mnt/nfs-data/users/yi/audio_phenomena_expansion_20260907`. These paths are remote filesystem references, not links to local files. Reproduction requires access to each manifest-bound source and compliance with that source’s access conditions; this thesis does not redistribute restricted recordings or turn file possession into redistribution permission.

A.2 Completed classification tables

The following entry points retain full lattices rather than only the selected comparisons displayed in the main text. Within `artifacts/audio_phenomena_expansion_20260907/`:

Exact10. `results/presentation_sdrfh_10s_v2/all_31x5_numeric_results.csv` contains the 31-subset, five-cap numeric presentation.

`results/presentation_sdrfh_10s_v2/RESULTS_SDRFH_10S_EN.md` provides the interpretation; `UNCERTAINTY_SDRFH_10S_RESULTS_EN.md` documents the separate conditional and source-composition uncertainty analyses.

Equal60. `results/presentation_equal60_v6_v1/primary_macro_overview_255.csv` contains all 255 subsets. `primary_matched_sc_deltas.csv` in that directory retains matched SC increments, including negative values.

`primary_source_endpoints_all_arms.csv` supplies source endpoints, and `training_row_group_ranges.csv` describes actual training support. Files prefixed `diagnostic_all_cap_` preserve the separately labelled diagnostic modes. The directory's `COMMIT.json` binds the presentation products; it does not replace the model-level numerical audit.

The earlier four-family figures are `artifacts/source_diversity_expansion_20260905/results/verified_presentation/10s_15x5_auc_ba.png` and the corresponding `30s_15x5_auc_ba.png`. Their different cohorts and feature/evaluation versions must be preserved in any cross-table comparison.

A.3 Measurement evidence and scope of audits

The external BC report is `artifacts/audio_phenomena_expansion_20260907/results/bc_reserved_verified_report_v2/REPORT_EN.md`, with its retained `REPORT.tex` and rendered `output/pdf/bc_reserved_verified_closeout_en.pdf` under the expansion root. Producer completion, independent numerical reconstruction and the scientific admission decision are separate evidence objects. Earlier failed audit records remain part of the history; the corrected storage-layout replay does not erase the original failure or change scientific tolerances.

The Native30 FHSC integrity report is `artifacts/audio_phenomena_expansion_20260907/audit/native30_fhsc_committed_independent_audit_v1/REPORT_EN.md`. Its file, schema, lineage and missingness checks must not be described as independent re-extraction of every waveform descriptor. Conversely, a unit test of a synthetic fixture is not evidence that the actual cohort completed. Each claim in the thesis uses the audit level applicable to that claim.

A.4 Execution and interpretation invariants

Reproduction follows the dependency graph, not directory modification times: verify source/view identities; verify frozen inference evidence; verify per-recording measurements; assemble matched feature rows; verify the fixed split/cap schedule; fit under a separate frozen evaluation contract; replay predictions and endpoint aggregation; then render tables. An experiment `COMMIT.json` denotes its immutable product manifest, not necessarily a Git commit. A metadata schedule does not authorize fitting and is not a count of completed models.

The CPU numerical path retains its frozen runtime and code dependencies. Neural inference in this project was restricted to RTX 5090 hardware; deterministic extraction and linear classification do not require a neural GPU call. Failed attempts, scientific nulls and processing errors are kept distinct. Resumption verifies existing products and does not replace failed values by zero, silently change a split, or select a more favourable window.

Only training rows determine imputation, scaling and fitted coefficients. Source/group weights, ridge penalty 10 and threshold 0.5 remain fixed. Reproduction must retain all modes and omissions, preserve transitive group purges, and avoid pooling raw scores from different fitted models into an unqualified global AUC. Ordinary-group results, held-Human-source results and held-generator results answer different questions.

A.5 Document build and completed evaluation bindings

The working master is `interpretable_audio_thesis_working_v7_20260912_en.tex`. From its containing directory, the retained build command is

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \  
interpretable_audio_thesis_working_v7_20260912_en.tex
```

Successful compilation is necessary but not sufficient: final acceptance also requires resolved citations, traceable tables and visual inspection of every page. The seven-family evaluation COMMIT SHA-256 is

`d1c16af22fa13b97fb2c9e69e6debc41b71c1d432cec86d1da035c06f3dd5efb`. The report COMMIT SHA-256 is

`fecd7bef116dd7bfa0c3cfa8307c508be4fcb58acfffdc447011c3606c7ddb46`. The completed BC, expanded YuE2 and Native60 transfer sections now provide their terminal report/evaluation bindings and local result paths. The first two report commits are respectively `ad5223a5f1f0fb66...` and `b7934f2c649e5d0f...`; complete hashes are printed in those sections. The Native60 prediction COMMIT is `34782ccfa467d530...`. Publication reconciliation, including redistribution restrictions for source audio, is separate from experimental completion and remains in progress. No prospective fit count is substituted for completion evidence.

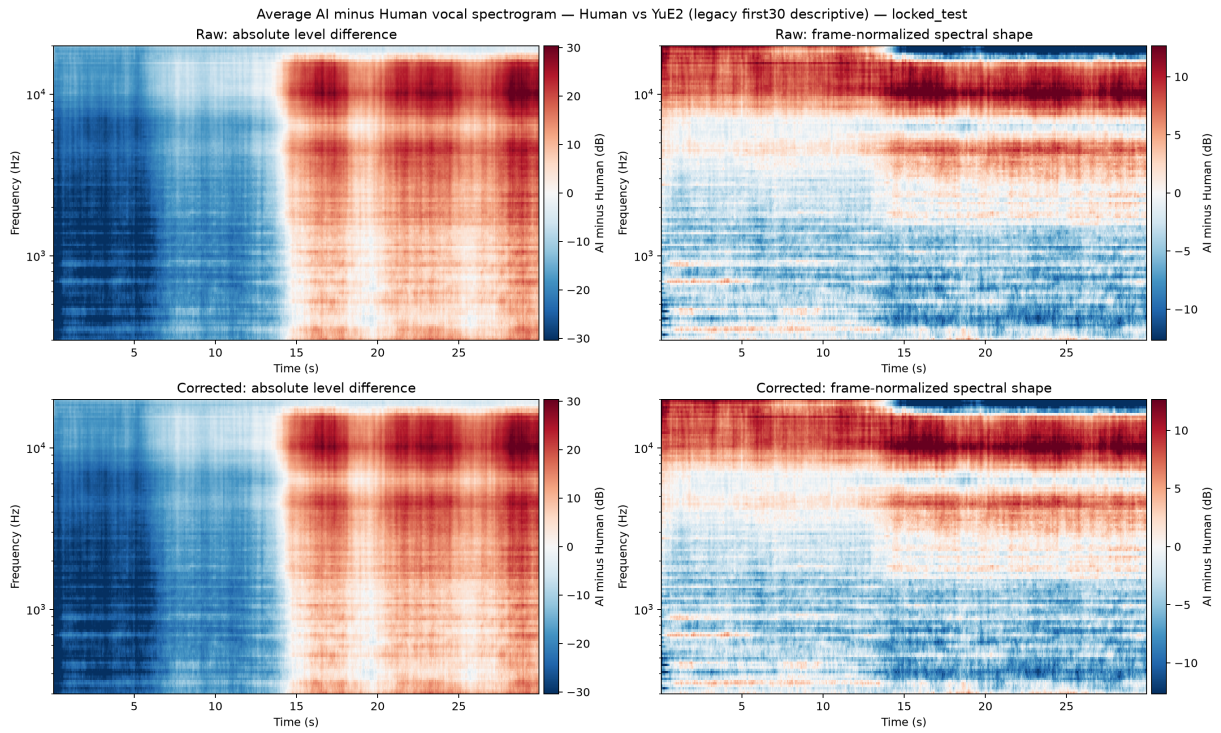


Figure 1: YuE2 minus human mean vocal spectrograms. Red indicates greater AI energy or relative band level; blue indicates the reverse. Raw and externally corrected panels retain broadly similar contrasts. The two classes are not waveform- or phrase-aligned. This is descriptive visualization, not a classifier test statistic or evidence of intrinsic AI causation.

B Audio preservation and public delivery audit

This appendix updates delivery status without changing experimental populations, feature definitions, fitted models, thresholds or numerical findings. Earlier pending-status statements in preserved artifacts describe their own checkpoints. Publication, historical provenance and experimental validity are distinct claims.

B.1 Fixed public snapshot

The public archive is [EZMONYI/music-ai-human-test-audio](https://ezmony.com/music-ai-human-test-audio). At the fixed revision

b5231d23713673e90c65e70767485ef5aaf70a56

it contains 42,011 direct audio/MIDI paths and 187,396,624,629 logical bytes. Archive members are excluded. These are file versions, not independent songs: originals, standardized excerpts and separated stems may represent the same recording. The subsequent dataset-card revision changes documentation, not audio.

Ten historical stem supplements passed anonymous fixed-revision acceptance. The tests compare the exact public manifest and notice and every remote audio hash and size; they do not redownload every waveform. Source cards are checked where required by the source-specific verifier.

Accepted supplement	Unique objects
ACE-Step/HeartMuLa vocals	1,958
AIME generated-music vocals	5,000
MAESTRO stems	2,400
MagnaTagATune stems	2,000
DiffRhythm pilot stems	400
Ishizaka stems	312
Additional Suno stems	1,300
Additional ACE-Step/HeartMuLa stems	600
Additional Udio stems	2,000
Per-recording-license-screened FMA stems	452
Total	16,422

The total is 47,275,922,568 bytes. Pilot publication does not promote pilot observations to primary evaluation data. FMA stems retain individual artist, title and exact license URL; there is no blanket archive license.

B.2 Coverage boundaries and local preservation

Of 31,378 historical-hash-verified stem objects, 16,422 match the public snapshot and 14,956 do not (50,814,442,064 unmatched bytes). A separate current-only inventory contains 27,174 objects: two match and 27,172 do not (68,652,547,568 unmatched bytes). These scopes overlap and must not be summed. Current file hashes cannot retroactively establish which bytes an earlier experiment read. There are also 126 unanchored historical references; these are not established missing-file findings. Snapshot matching does not inspect archive contents or private backups and does not grant redistribution rights.

The frozen 14,244-object processed-input scope has preservation coverage through 966 older public-snapshot matches, 8,794 accepted public supplement objects and 4,484 locally archived rights-held objects. The local TAR is 9,217,576,960 bytes; all 4,484 audio members passed streaming SHA-256 verification. It is not publicly released. This closes that preservation scope only, not whole-project delivery. Other unresolved source licenses and historical input scopes remain open.

B.3 Code, output and audit references

The code repository is [ezmonyi/ai-music-detection](https://github.com/ezmonyi/ai-music-detection). Within its `datasets/` directory, the versioned evidence includes `public_audio_snapshot_20260913_v3`, `accepted_historical_stem_coverage_v2`, `stem_public_snapshot_match_v3`, and `processed_delivery_reconciliation_v2`. These retain manifest digests, publication revisions, exact object matches and limitations; source-specific publication directories retain acceptance receipts. The corresponding scripts are under `current/yue2_500_extension_20260911/`, including `snapshot_public_audio_inventory_v1.py`, `reconcile_accepted_historical_stems_v2.py`, and `match_stems_public_snapshot_v3.py`.

The local report root is `/Users/yi/Documents/report/music_ai_detection_20260912/`. It preserves report versions, receipts, intermediate analysis and the private archive; `START_HERE_EN.md` is its navigation entry point. Preserving an artifact is not proof that its scientific claim is valid. The evaluation limitations discussed in the main text remain unchanged.